ASSESSMENT 2

Deadline:	Hand in by midnight 23 September.
Evaluation:	**20% of your final course grade.**
Late Submission:	Refer to the course guide.
Work	This assignment may be done in **pairs** or **individually**.
Purpose:	Learning outcomes 1 to 4 from the course outline.

PROJECT DESCRIPTION

This project is about forecasting and solving a real-world problem, and to a large degree, about Gradient Boosting too.

You will be provided with a target variable which you need to model using a range of input data and engineered features. The target variable is New Zealand inflation expressed as a time-series. You will be asked perform training and testing using a specific experimental design, and your best test score will be entered into a in-class competition. The in-class competition will be conducted using the gradient boosting algorithm implementation from the class notebook which you will be allowed to tweak and optimise for best performance on this dataset. You will be told which sources of input data for predicting the inflation you're permitted to use and you will need to acquire this data yourself. The project has 5 tasks as the previous assignment.

HAND-IN:

Make sure that all the outputs are embedded in your notebook and submit both the notebook and the html version, plus your final dataset as a csv file. Do not do data-dumps unless it is your final training dataset. **Again, keep your notebooks tidy and readable. Make sure you provide analyses of your outputs/findings/graphs, otherwise penalties will be applied.**

Use a project template provided on Stream and describe at the top all the steps you have performed – this is important.

Aim to submit just 1 notebook please but maximum of 2 if you must in order to include AWS components. Submit an HTML version + PDF versions of your work too please.

TASK 1: GRADIENT BOOSTING IMPLEMENTATION (25 MARKS)

Complete the following tasks by using the provided target variable together with your chosen features.

1. Implement Gradient Boosting Algorithm: (5 marks)
    a. Correctly implement the simplified Gradient Boosting algorithm for regression **using class code** from the relevant notebook and demonstrate that it works. <u>You must make sure you work with the Gradient Boosting code backbone from the class. Completely different implementations of GB having a significant departure from the class code will not count in this component.</u>
2. Adjust the Loss Function: (5 marks)
    a. Modify the algorithm code so that it can work with alternative loss functions. Experiment with at least TWO different loss functions and evaluate their effect on both the training convergence and test error.
3. Adjust Weak Learner Type: (5 marks)
    a. Conduct experiments with different weak learner types within the algorithm. Try at least TWO different ones. They can be from scikit-learn, or otherwise.
    b. Observe how the choice of weak learner impacts the overall performance of the model and report and discuss this clearly.
4. Modify the Learning Rate Schedule: (5 Marks)
    a. Devise and implement a dynamic learning rate schedule where the learning rate decreases as the number of iterations increases. Explore different schedules and provide a set of experiments as well as a discussion to report on your findings seeking to analyse how different learning rate schedules affect model convergence, overfitting, and final performance. Discuss the trade-offs between starting with a higher learning rate versus a more conservative fixed learning rate.
5. Implement Gradient-Based Early Stopping: (5 Marks)
    a. Devise and implement an early stopping mechanism which stops the training based on some property of the loss function perhaps in conjunction with a validation sample.

TASK 2: GRADIENT BOOSTING IN-CLASS FORECASTING COMPETITION (35 MARKS)

We will be conducting an in-class competition to determine the student who achieves the best solution using **our class implementation of Gradient Boosting**. You must use your implementation of the algorithm based on the class notebook which potentially features some of your modifications/enhancements. Third party implementations from scikit-learn or other modules **are not valid for this**.

<u>RULES</u>:

- Reporting evaluation metric: Mean Squared Error (MSE)
- Permitted feature dataset sources: Reserve Bank of NZ, Statistics NZ, MBIE. If you wish to use an additional dataset, then you must seek my permission and be prepared to share this dataset with the rest of the class for fairness. The last opportunity to do this will be one week before submission deadline.
- You must also take care that your chosen variables do not constitute 'data leakage' by becoming known/available only after inflation figures are actually released – investigate this and discuss in the notebook how you have ensured this does not happen.
- You must report your test dataset MSE score using the expanded window approach.
- **You must not use any of the test data for training**. This will disqualify you from the competition.
- You can use various data imputation methods for missing values but the imputation **must not** include test data as an input.
- You must show the dataframe in your notebook that contains the full dataset and you must submit this dataset in a csv format as well. You must also make sure that you embed into your notebook the MSE score and the performance plots of your best scoring model which you reported in the class Google Sheet leaderboard.

1. Competition Score Calculation: (20 marks)

   a. The competition score for each student's (or pair of students) submission will be based on their model's performance on a test dataset MSE score which must be generated using scikit-learn
   b. Each student must enter their best score on a Google Sheet document that will be provided
   c. Evidence in the form of graphs and the associated MSE embedded in the notebook demonstrating their best model must be depicted and highlighted in the notebook for the score reported on the Google Sheet
   d. The winning student(s) will receive 20 marks for obtaining the top-performing (lowest MSE) solution
   e. The student with the next best solution will be awarded 17 marks. Subsequent rankings will receive one less mark than the preceding position, with the minimum being 10 marks for a working model with some MSE score.
   f. Your first MSE score must be entered by **September 5** into the class competition Google Sheet by this date, otherwise 5 marks will be deducted.
2. Feature engineering, algorithm customisation, thoroughness in experimentation and reporting (15 marks + potentially 5 bonus marks for exceptional work)
   a. Experiment with different combinations of customisations of Gradient Boosting from Task 1 to optimise its performance with explanations and interpretations of what might be occurring.
   b. Proactiveness in data acquisition and experimentation with different data inputs.
   c. Creativity in conducting feature engineering/selection to identify the most relevant and informative features for the forecasting task.
   d. Overall reporting, clarity and interpretations on how the performance of the models change with different adjustments, feature subsets problem formulations.
   e. Use of a variety of performance metrics: MAE, MAPE and others
   f. Can you make a forecast for the inflation in Q3?

TASK 3: EXPERIMENTATION USING A RANGE OF ALGORITHMS (10 MARKS)

1. Compare your best algorithm customisation with the performance of XGBoost, LightGBM, CatBoost and the scikit-learn implementation and report your findings
2. Explore a range of other different algorithms and their performance
3. Demonstrate hyperparameter tuning where necessary
4. Document MSE performance over time with various tweaks and strategies
   a. Create visualizations, such as line plots, to display the MSE trend of various experiments.

TASK 4: AWS TECHNOLOGY USAGE (15 MARKS)

In this task, demonstrate your usage of AWS technologies to build solution to the forecasting problem. Use any AWS services covered thus far in the course.

1. AWS Demonstration: (10 marks)

In this task, you will showcase your proficiency in utilising AWS technologies covered thus far in the course, including AWS forecasting technologies. Alongside forecasting capabilities, the focus will also be on applying AWS services to enhance machine learning workflows, leveraging cloud computing capabilities for scalable and efficient data processing (even when the datasets are small like this one). The key AWS technologies to consider demonstrating are as follows:

1. Amazon SageMaker:
2. Amazon S3 (Simple Storage Service):
3. AWS Glue:
4. Hyperparameter Tuning:
5. Amazon SageMaker Autopilot:
6. AWS Forecast
7. Other…

However, not all the above need to be used.

2. AWS Course Completion: (5 marks)

Include a screenshot of your AWS Academy course progression and embedding the image into the notebook, showing how far you have progressed in your AWS Academy courses. Without this, zero marks will be awarded for Task 4.

TASK 5: FILLING OUT THE JUPYTER NOTEBOOK TEMPLATE (15 MARKS)

In the top cell of your notebook, you must complete a checklist/preamble in markdown that confirms you have addressed each of the major components of the assignment. For every requirement listed, you must provide a working internal hyperlink that directs the marker to the exact section of your notebook where you have completed the corresponding work. This will allow us to verify your work efficiently and is essential to the marking process. You are expected to be explicit and accurate with your links. Incomplete or vague references will result in penalties.

This part of the assignment is not just administrative — it is designed to prompt you to reflect critically on the completeness and quality of your work. This checklist represents your own evaluation of how well you have met the task objectives, and is an important part of the learning outcomes. Being able to map your work to specific deliverables is a key skill in both academic and professional settings, and the process of doing so supports your ability to evaluate your own progress and performance. You should approach this section with care and attention, and treat it as a formal declaration of what you have done, what you are submitting, and how it aligns with the assessment criteria. Use the notebook markdown and internal linking syntax properly, and ensure that the linked content is clearly presented and corresponds precisely to the task described.

It goes without saying, that using GenAI to write assignment solutions for you is not acceptable and constitutes academic dishonesty and cheating.